Common Network Attacks

1.0.1.1

Overview of Network Attacks

As you have learned, there are many types of malware that hackers can use. But these are not the only ways that they can attack a network, or even an organization.

When malware is detected and installed, the payload can be used to cause a variety of network related attacks.

To mitigate attacks, it is useful to understand the types of attacks. By categorizing network attacks, it is possible to address types of attacks rather than individual attacks.

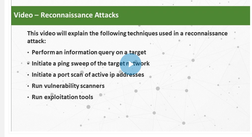
Networks are susceptible to the following types of attacks:

- Reconnaissance Attacks
- Access Attacks
- DoS Attacks

1.0.1.2

Video – Reconnaissance Attacks

Click Play in the figure to view a video about reconnaissance attacks.



1.0.1.3

Reconnaissance Attacks

Reconnaissance is information gathering. It is analogous to a thief surveying a neighborhood by going door-to-door pretending to sell something. What the thief is actually doing is looking for vulnerable homes to break into, such as unoccupied residences, residences with easy-to-open doors or windows, and those residences without security systems or security cameras.

Threat actors use reconnaissance (or recon) attacks to do unauthorized discovery and mapping of systems, services, or vulnerabilities. Recon attacks precede access or DoS attacks.

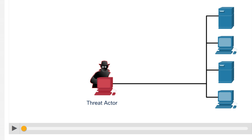
Some of the techniques used by malicious threat actors to conduct reconnaissance attacks are described in the table.

Technique	Description
Perform an information query of a target	The threat actor is looking for initial information about a target. Various tools can be used, including the Google search, nslookup, and the dig command-line tool.
Initiate a ping sweep of the target network	The information query usually reveals the target's network address. The threat actor can now initiate a ping sweep to determine which IP addresses are active.
Initiate a port scan of active IP addresses	This is used to determine which ports or services are available. Examples of port scanners include Nmap, SuperScan, Angry IP Scanner, and NetScanTools.
Run vulnerability scanners	This is used to query the targeted hosts to determine the type and version of the application and operating system that is running on the host. Examples of tools include Nessus, Secuna PSI, Core Impact, Nessus, ISS, SAINT, and Open VAS.
Run exploitation tools	The threat actor now attempts to discover vulnerable services that can be exploited. A variety of vulnerability exploitation tools exist including Metasploit, Core Impact, Sqlmap, Social Engineer Toolkit, and Netsparker.

Click each button to view the progress of a reconnaissance attack from information query, to ping sweep, to port scan.

Information Query | Performing Ping Sweep | Performing Port Scans

Click Play in the figure to view an animation of a threat actor using the whois command to find information about a target.



1.0.1.4

Video – Access and Social Engineering Attacks

Click Play in the figure to view a video about access and social engineering attacks.

1.0.1.5

Access Attacks

Access attacks exploit known vulnerabilities in authentication services, FTP services, and web services. The purpose of these types of attacks is to gain entry to web accounts, confidential databases, and other sensitive information.

Threat actors use access attacks on network devices and computers to retrieve data, gain access, or to escalate access privileges to administrator status.

Password Attacks

In a password attack, the threat actor attempts to discover critical system passwords using various methods. Password attacks are very common and can be launched using a variety of password cracking tools.

Spoofing Attacks

In spoofing attacks, the threat actor device attempts to pose as another device by falsifying data. Common spoofing attacks include IP spoofing, MAC spoofing, and DHCP spoofing. These spoofing attacks will be discussed in more detail later in this module.

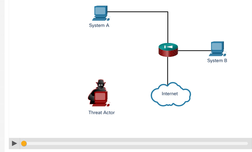
Other Access attacks include:

- Trust exploitations
- Port redirections
- Man-in-the-middle attacks
- Buffer overflow attacks

Click each button to view an illustration and explanation of these access attacks.

Trust Exploitation Example | Port Redirection Example | Man-in-the-Middle Attack Example | Buffer Overflow Attack

In a trust exploitation attack, a threat actor uses unauthorized privileges to gain access to a system, possibly compromising the target. Click Play in the figure to view an example of trust exploitation.



1.0.1.6

Social Engineering Attacks

Social engineering is an access attack that attempts to manipulate individuals into performing actions or divulging confidential information. Some social engineering techniques are performed in-person while others may use the telephone or Internet.

Social engineers often rely on people's willingness to be helpful. They also prey on people's weaknesses. For example, a threat actor could call an authorized employee with a sob story or promise to fix a network access. The threat actor could appeal to the employee's vanity, invoke authority, using name-dropping techniques, or appeal to the employee's greed.

Information about social engineering techniques is shown in the table.

Social Engineering Attack	Description
Pretexting	A threat actor pretends to need personal or financial data to confirm the identity of the recipient.
Phishing	A threat actor sends fraudulent email which is disguised as being from a legitimate, trusted source to trick the recipient into installing malware on their device, or to share personal or financial information.
Spear phishing	A threat actor creates a targeted phishing attack tailored for a specific individual or organization.
Spam	Also known as junk mail, this is unsolicited email which often contains harmful links, malware, or deceptive content.
Something for Something	Sometimes called "Quid pro quo", this is when a threat actor requests personal information from a party in exchange for something such as a gift.
Baiting	A threat actor leaves a malware infected flash drive in a public location. A victim finds the drive and unsuspectingly inserts it into their laptop, unintentionally installing malware.
Impersonation	The type of attack is where a threat actor pretends to be someone they are not to gain the trust of a victim.
Tailgating	This is where a threat actor quickly follows an authorized person into a secure location to gain access to a secure area.
Shoulder surfing	This is where a threat actor inconspicuously looks over someone's shoulder to steal their passwords or other information.
Dumpster diving	This is where a threat actor rummages through trash bins to discover confidential documents.

The Social Engineering Toolkit (SET) was designed to help skilled ethical hackers and other network security professionals create social engineering attacks to test their own networks.

Enterprises must educate their users about the risks of social engineering, and develop strategies to validate identities over the phone, via email, or in person.

The figure shows recommended practices that should be followed by all users.

Recommended Social Engineering Protection Practices

1.0.1.7

Lab - Social Engineering

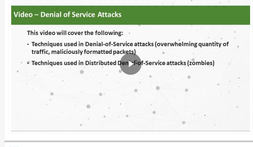
In this lab, you will research examples of social engineering and identify ways to recognize and prevent it.

Social Engineering

1.0.1.8

Video - Denial of Service Attacks

Click Play in the figure to view a video about denial of service attacks.



1.0.1.9

DoS and DDoS Attacks

A Denial of Service (DoS) attack creates some sort of interruption of network services to users, devices, or applications. There are two major types of DoS attacks:

- **Overwhelming Quantity of Traffic** – The threat actor sends an enormous quantity of data at a rate that the network, host, or application cannot handle. This causes transmission and response times to slow down. It can also crash a device or service.
- **Maliciously Formatted Packets** – The threat actor sends a maliciously formatted packet to a host or application and the receiver is unable to handle it. This causes the receiving device to run very slowly or crash.

Click each button for an illustration and explanation of DoS and DDoS attacks.

DoS Attack | DDoS Attack

DoS attacks are a major risk because they interrupt communication and cause significant loss of time and money. These attacks are relatively simple to conduct, even by an unskilled threat actor.

Click Play in the figure to view the animation of a DoS attack.

1.0.1.10

Check Your Understanding - Common Network Attacks

Check your understanding of common network attacks by choosing the BEST classification for each attack type.

1. What type of attack is tailgating?
   - Reconnaissance
   - Access
   - DoS
   - Social Engineering
   Social Engineering

2. What type of attack is a password attack?
   - Reconnaissance
   - Access
   - DoS
   - Social Engineering
   Access

3. What type of attack is port scanning?
   - Reconnaissance
   - Access
   - DoS
   - Social Engineering
   Reconnaissance

4. What type of attack is man-in-the-middle?
   - Reconnaissance
   - Access
   - DoS
   - Social Engineering
   Access

5. What type of attack is address spoofing?
   - Reconnaissance
   - Access
   - DoS
   - Social Engineering
   Access

Check | Show Me | Reset